

100 Days of Linux - Day 030

Title: Common Linux Permission Values (644, 755, 600, 700)

Today's Goal

Understand the most commonly used Linux permission values and know when to use each one in real projects.

Why This Matters

Choosing the correct permissions protects sensitive files while allowing programs and users to work correctly. These values appear constantly in software development, DevOps, and data engineering.

Permission Values of the Day

644 = Owner: read/write, Group: read, Others: read (common for text files)

755 = Owner: read/write/execute, Group: read/execute, Others: read/execute (common for scripts and directories)

600 = Owner: read/write only (private files)

700 = Owner: read/write/execute only (private directories and scripts)

Examples:

```
chmod 644 notes.txt
```

```
chmod 755 pipeline.sh
```

```
chmod 600 secrets.txt
```

```
chmod 700 private_folder
```

Beginner Lesson

Permission numbers are simply combinations of read (4), write (2), and execute (1). Add them together for each permission group. Today you'll practice using the four values you will see most often.

Practice Questions

1. Create a file named notes.txt and set its permissions to 644. Verify using ls -l.
2. Create a shell script named pipeline.sh and set its permissions to 755.
3. Create a file named secrets.txt and set its permissions to 600.
4. Create a directory named private_data and set its permissions to 700. Verify the result.
5. Display your current working directory and list all files with detailed permissions.

Hints

Always run ls -l after chmod to confirm the permissions were applied correctly.

Mini Challenge

Create a data_project folder containing script.sh, config.env, report.txt, and private_notes.txt. Apply 755 to the script, 600 to config.env and private_notes.txt, and 644 to report.txt. Verify every file with ls -l.

Common Mistakes

Using 755 on confidential files or forgetting execute permission for scripts that need to run.

Did You Know?

Many Git repositories store shell scripts with 755 so they can be executed immediately after cloning.

Pro Tip

Memorize these four values: 644 (files), 755 (scripts/directories), 600 (private files), and 700 (private directories).

Answer Key (Instructor Only)

Q1: touch notes.txt && chmod 644 notes.txt && ls -l notes.txt

Q2: touch pipeline.sh && chmod 755 pipeline.sh && ls -l
pipeline.sh

Q3: touch secrets.txt && chmod 600 secrets.txt && ls -l secrets.txt

Q4: mkdir private_data && chmod 700 private_data && ls -ld
private_data

Q5: pwd && ls -l

Homework

Create a small practice project and apply 644, 755, 600, and 700 to appropriate files until you can choose the correct permission without looking at your notes.

Next Day Preview

Tomorrow you'll begin the Text Processing module with the sort command and learn how to organize data alphabetically and numerically.